

PROBLEM SET 5

This assignment is for your practice and will not be graded.

1. Consider the following version of the perfect information War of Attrition. At the beginning of each period t , $n > 1$ identical firms simultaneously decide whether to exit or stay in a market for a homogeneous good. Any firm that exists earns zero from that date on (reentry is prohibited for simplicity). The cost of staying in the market is $f > 0$ in each period and the cost of exiting is zero. Each firm observes whether or not its rival exits prior to setting its price, and marginal costs are constant. Following the exit stage, one of three situations may obtain. (1) There may be no firms left in the industry. (2) There may be a single firm left that earns $\Pi^M - f > 0$ from that date on. (3) There may be more than one firm left in the industry, in which case each incurs net loss of $-f$ in the current period (Bertrand competition). Let the common discount factor be $\delta < 1$, and suppose that all costs and benefits occur at the beginning of each period.
 - (a) Suppose that there are two firms in the industry in period zero. Firm 2 randomizes over its exit decision in each period (including zero). In particular, it leaves the industry with probability P in period t if firm 1 has not exited prior to that time. Find the value for P that leaves firm 1 indifferent between exiting or staying in the industry in a given period.
 - (b) Characterize the symmetric-mixed strategy equilibrium of this duopoly game. Characterize at least one asymmetric equilibrium that also exists.
 - (c) Now suppose that there are $n > 2$ identical firms in the industry in period zero. What is the formula for $P^*(n)$, the probability of exit by a given firm in period zero under the symmetric mixed-strategy equilibrium? Do you need to worry about subgames in which $M = 2, \dots, N-1$ firms remain in the industry when computing $P^*(n)$? Why/why not?
 - (d) Return to the duopoly model, and consider the potential for collusion. It is not credible in the model to threaten to punish deviations with deterministic marginal cost pricing forever since deterministically staying in the market and charging marginal cost is not an equilibrium. However, it is credible to threaten to punish deviations with a War of Attrition. Find the value $\underline{\delta}$ of the discount factor such that perfect symmetric collusion can be sustained iff $\delta \geq \underline{\delta}$ assuming that $0.5\Pi^M - f > 0$. Compute the value for $\underline{\delta}$ when $f = 0$. What do you conclude about the ability to support collusion in the presence of fixed costs relative to their absence?
2. Consider a variation of the Rubinstein-Stahl perfect information infinite horizon bargaining game between two parties $i = 1, 2$. At the beginning of each period t Nature selects one of the players to be the proposer. In particular, player 1 is the proposer in each period with probability α and player 2 is proposer in each period with probability $1 - \alpha$. The proposer in a given period suggests a division $(x, 1 - x)$, where x is a player 1's share of the surplus. The other player may either accept the proposal or reject it. If he rejects, then the game moves to the next period where the process begins again with a new move by Nature. Let $\delta_i \in (0, 1)$ be player i 's discount factor.
 - (a) What is the unique SPNE of this game? (You do not have to prove uniqueness).
 - (b) What are the players ex ante expected equilibrium payoffs in each period (before the proposer has been selected).
 - (c) Calculate the limit as α goes to 1 of the payoffs you found in part b). Briefly interpret the limit.
3. Consider the following asymmetric-information model of Bertrand duopoly with differentiated products. Demand for firm i is $q_i(p_i, p_j) = a - p_i - b_i p_j$. Costs are zero for both firms. The sensitivity of firm i 's demand to firm j 's price is either high or low. That is $b_i = b_H$ with probability θ or $b_i = b_L$ with probability $1 - \theta$ where $b_H > b_L > 0$. Each firm knows their own b_i but not the competitor's.
 - (a) What are the action spaces, type spaces, beliefs and payoff functions in this game?
 - (b) Derive the symmetric pure-strategy Bayesian Nash equilibrium.

4. Find all pure-strategies BNE in the following static Bayesian game:

1. Nature determines whether the payoffs are as in Game 1 or as in Game 2 depicted in Figure 1 and 2 respectively. Each game is equally likely.
2. Player 1 learns whether nature has drawn Game 1 or Game 2 but player 2 does not.
3. Player 1 chooses either U or D and player 2 simultaneously chooses either L or R.

		G 1 P 2				G 2 P 2	
		<i>L</i>	<i>R</i>			<i>L</i>	<i>R</i>
P 1	<i>U</i>	1, 1	0, 0	P 1	<i>U</i>	0, 0	0, 0
	<i>D</i>	0, 0	0, 0		<i>D</i>	0, 0	2, 2